Submission Highlights

Data Quality for Derived Preference Graphs: Construction Sensitivity and Repair Outcomes in Multi-Ranker Retrieval

- Preference graphs built from heterogeneous retrieval scores are treated as derived data artifacts, not as trusted raw inputs to graph repair.
- A compact seven-dimension data-quality audit framework separates provenance, calibration, vote semantics, conflict structure, repair quality, downstream utility, and reproducibility.
- Raw score margins are strongly scale dominated: BM25's conditional mean edge-weight share is 0.988 under raw construction but 0.512 after per-query, per-ranker normalization.
- Repair changes graph structure and, under genuine $P > k$ evaluation, can change top-k membership, but no repaired-versus-unrepaired nDCG family in this study survives the primary Holm corrections.
- Exact SCIP repair improves the structural objective without revealing a reliable retrieval gain, yielding a practitioner-facing conclusion: normalization, vote construction, pooling, and evaluation cutoffs must be reported as first-class data-quality choices.